**BVS1 · Winter Semester 2026/2027 · TH Köln · ICCT**

# Handout — Editing Source Files in Ubuntu — nano and Vim

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You only need one editor for BVS1. If you have not used a terminal editor before, **nano is usually the quickest way to get started**. Use **Vim** if you already know it or want to learn modal editing.

The editor is not part of the assessment. The goal is simply to edit source files reliably.

## nano — quickest start

Open a file:

```
nano main.c
```

Nano immediately lets you type and edit text. At the bottom of the screen, `^` means **Ctrl**. For example, `^O` means **Ctrl-O**.

| Action | Key |
|---|---|
| Save | **Ctrl-O**, then **Enter** |
| Exit | **Ctrl-X** |
| Search | **Ctrl-W** |
| Go to line | **Ctrl-_** |
| Cut current line | **Ctrl-K** |
| Paste last cut text | **Ctrl-U** |
| Undo | **Alt-U** |
| Redo | **Alt-E** |
| Help | **Ctrl-G** |

A safe everyday sequence is:

```
nano main.c
    ↓
edit
    ↓
Ctrl-O, Enter
    ↓
Ctrl-X
```

If Nano asks whether to save modified content when you leave, answer deliberately. Do not press keys blindly.

## Vim — minimum survival kit

Open a file:

```
vim main.c
```

Vim is **modal**. The two modes you need first are:

- **Normal mode** — navigation and commands;
- **Insert mode** — typing text.

Vim starts in Normal mode.

### Edit, save, leave

| Action | Key / command |
|---|---|
| Enter Insert mode | **i** |
| Return to Normal mode | **Esc** |
| Save | **:w** then **Enter** |
| Quit | **:q** then **Enter** |
| Save and quit | **:wq** then **Enter** |
| Quit and discard changes | **:q!** then **Enter** |

If you are unsure which mode you are in, press **Esc**. Pressing it more than once is harmless.

### Useful Normal-mode commands

| Action | Key / command |
|---|---|
| Move | arrow keys or **h j k l** |
| Beginning / end of line | **0** / **$** |
| First / last line | **gg** / **G** |
| Delete current line | **dd** |
| Copy current line | **yy** |
| Paste after cursor | **p** |
| Delete one character | **x** |
| Undo | **u** |
| Redo | **Ctrl-R** |
| Search forward | **/text**, then **Enter** |
| Next / previous match | **n** / **N** |
| Show line numbers | **:set number**, then **Enter** |

## Vim emergency exits

If Vim appears to be doing something unexpected:

```
Esc
```

Then choose one:

```
:wq    save and leave
:q!    leave without saving
```

and press **Enter**.

If typed letters are becoming commands instead of text, press **i** to enter Insert mode.

If commands such as `:wq` are becoming part of the file, press **Esc** first.

## Editing BVS1 source

Typical workflow:

```
cd ~/teaching-bvs1-code
nano main.c
```

or:

```
cd ~/teaching-bvs1-code
vim main.c
```

After saving, inspect what changed before rebuilding:

```
git diff
```

Then build or run the experiment requested by the current lecture, exercise, or checkpoint.

## Remember

- Pick **one** editor and become comfortable with its save/quit workflow.
- Save before building.
- Use `git diff` to check what you actually changed.
- In Vim, **Esc** gets you back to Normal mode.
- In Nano, the command hints at the bottom of the screen are there to help.